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INSTRUCTION MANUAL

ZymoBIOMICS™ Microbial Community DNA Standard II (Log Distribution) Catalog No. D6311

Highlights

- **Log abundance distribution:** assess detection limit of as low as DNA of three microbes.
- **Accurate composition:** cross-validated with multiple types of measurements.
- **Microbiomics QC:** ideal for quality control of microbiome measurements

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Satisfaction of all Zymo Research products is guaranteed. If you are dissatisfied with this product, please call 1-888-882-9682.

Note – Integrity of kit components is guaranteed for up to one year from date of purchase.

¹ If you have difficulty accessing the Certificate of Analysis with the link, please contact our tech support team at: 949-697-1190

Notes:
This product is for research use only and should only be used by trained professionals. It is not for use in diagnostic procedures. Some reagents included with this kit are irritants. Wear protective gloves and eye protection. Follow the safety guidelines and rules enacted by your research institution or facility.

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Product Contents

Product Name	D6311	Storage Temperature
ZymoBIOMICS™ Microbial Community DNA Standard II (Log Distribution)	220ng / 20µl	-20°C

Product Specifications

Source: genomic DNA of eight bacteria and two yeasts.

Reference genomes and 16S&18S rRNA genes:

<https://s3.amazonaws.com/zymo-files/BioPool/ZymoBIOMICS.STD.refseq.v2.zip>.

Storage solution: 10 mM Tris-HCl and 0.1 mM EDTA, pH 8.0.

DNA concentration: 11 ng/µl.

Impurity level: contain < 0.01% foreign microbial DNA.

Relative-abundance deviation in average: <30%.

Microbial composition: Table 1 shows the theoretical microbial composition of the standard.

The microbial composition of each lot was measured by shotgun metagenomic sequencing post mixing. The results (including the composition, impurities and abundance deviation) can be accessed through the Certificate of Analysis based on the lot number (printed on tube label) by the following link: <http://www.zymoresearch.com/microbiomics/microbial-standards/zymbiomics-microbial-community-standards> ¹.

Table 1: Microbial Composition

Species	Theoretical Composition (%)				
	Genomic DNA	16S Only ¹	16S & 18S ¹	Genome Copy ²	Cell Number ³
<i>Listeria monocytogenes</i>	89.1	95.9	91.9	94.8	94.9
<i>Pseudomonas aeruginosa</i>	8.9	2.8	2.7	4.2	4.2
<i>Bacillus subtilis</i>	0.89	1.2	1.1	0.7	0.7
<i>Saccharomyces cerevisiae</i>	0.89	NA	4.1	0.23	0.12
<i>Escherichia coli</i>	0.089	0.069	0.066	0.058	0.058
<i>Salmonella enterica</i>	0.089	0.07	0.067	0.059	0.059
<i>Lactobacillus fermentum</i>	0.0089	0.012	0.012	0.015	0.015
<i>Enterococcus faecalis</i>	0.00089	0.00067	0.00064	0.001	0.001
<i>Cryptococcus neoformans</i>	0.00089	NA	0.0014	0.00015	0.00007
<i>Staphylococcus aureus</i>	0.000089	0.0001	0.0001	0.0001	0.0001

¹ The theoretical composition in terms of 16S (or 16S & 18S) rRNA gene abundance was calculated from theoretical genomic DNA composition with the following formula: 16S/18S copy number = total genomic DNA (g) × unit conversion constant (bp/g) / genome size (bp) × 16S/18S copy number per genome. Use this as reference when performing 16S targeted sequencing.

² The theoretical composition in terms of genome copy number was calculated from theoretical genomic DNA composition with the following formula: genome copy number = total genomic DNA (g) × unit conversion constant (bp/g) / genome size (bp). Use this as reference when inferring microbial abundance from shotgun sequencing data based on read depth.

³ The theoretical composition in terms of cell number was calculated from theoretical genomic DNA composition with the following formula: genome copy number = total genomic DNA (g) × unit conversion constant (bp/g) / genome size (bp)/ploidy.

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Product Description

ZymoBIOMICS™ Microbial Community DNA Standard II (Log Distribution) is a mixture of genomic DNA of eight bacterial and two fungal strains. The microbial standard is accurately characterized and contains negligible impurity (< 0.01%). It was constructed by pooling DNA extracted from pure cultures of the ten microbial strains¹. The DNA from each pure culture was quantified before pooling. After mixing, the microbial composition was confirmed using NGS-based sequencing (Figure 1). This microbial standard can be used to assess the performance of microbiomics workflows and can also be used as a positive control for the routine QC purpose.

DNA samples were mixed to create log-distributed abundance (Table 1, Page 1), which allows the user to easily assess the detection limit of a microbiomics workflow. 1 µl of the standard (11 ng DNA) can be used to assess the detection limit of as low as the abundance of *Staphylococcus aureus* contained in the standard, which is 0.000089% by relative abundance or is equivalent to the amount of DNA from 3 cells. If desired, the standard can also be used to mix with human genomic DNA, e.g. Human HCT116 DKO DNA (Cat. No. D5014-1), to mimic a human microbiome sample.

Details regarding the ten microbial strains (including species name, genome size, ploidy, average GC content, 16S/18S copy number, phylogeny) can be found in Table 2 (Page 3). The 16S/18S rRNA sequences (FASTA format) and genomes (FASTA format) of these strains are available at: <https://s3.amazonaws.com/zymo-files/BioPool/ZymoBIOMICS.STD.refseq.v2.zip>. Feel free to contact us if you need help analyzing sequencing data generated from this standard².

Background on the Need for Microbiome Standards: Microbial composition profiling techniques powered by next-generation sequencing are becoming routine in microbiomics and metagenomics studies. It is well known that these analytical techniques can suffer from bias and errors in every step of the workflow, including DNA extraction, library preparation, sequencing and bioinformatics analysis. To assess the performance of different microbiomics workflows, there is an urgent need in the field for reliable reference materials, e.g. a mock microbial community with defined composition.

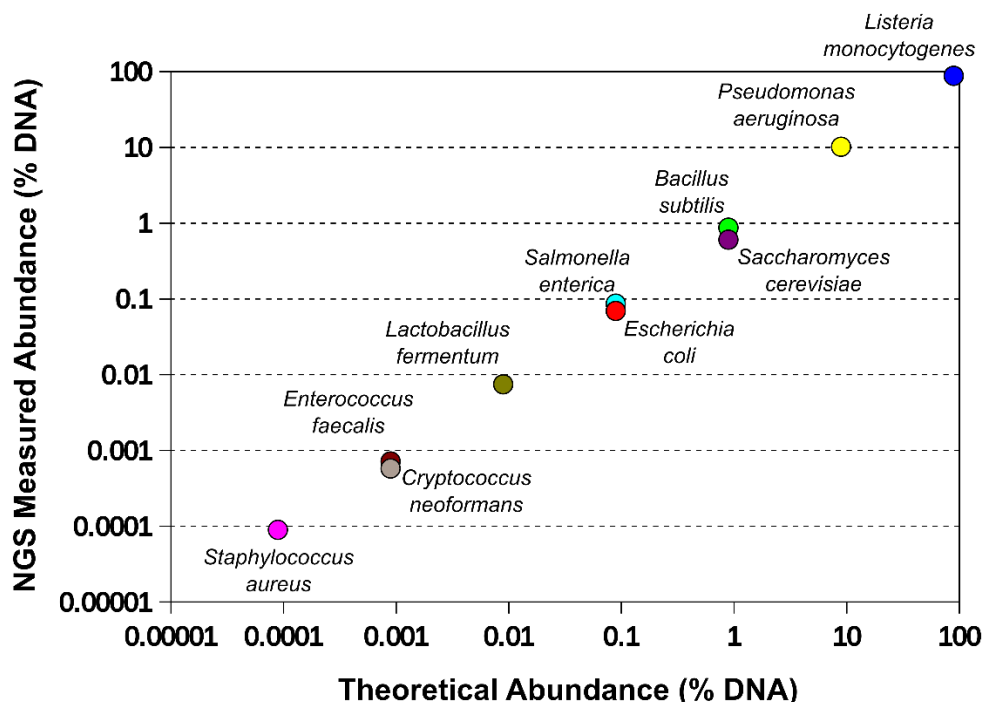


Figure 1. The microbial composition of the standard measured by NGS shotgun sequencing as compared to the defined composition. After mixing, the microbial composition of the standard was confirmed using deep Illumina shotgun sequencing. Briefly, library preparation was performed using an in-house protocol. Shotgun sequencing was performed using Illumina HiSeq™ or MiSeq™. Microbial abundance was estimated based on the number of reads that were mapped to reference genomes of the organisms.

Notes:

¹ Genomic DNA from each culture was extracted and quantified before mixing so this DNA standard was independent and not a direct derivative of the microbial version, ZymoBIOMICS™ Microbial Community Standard II (Staggered, Cellular Mix).

² We can use in-house pipelines to help assess the extent of bias in the sequencing data of this standard.

Notes:

¹ 18S rRNA gene copy numbers in a haploid genome of the two strains of *Saccharomyces cerevisiae* and *Cryptococcus neoformans* were estimated based on read depth information from mapping shotgun sequencing data.

Strain Information**Table 2: Strain Information**

Species	NRRL Accession NO.	Genome Size (Mb)	Ploidy	GC Content (%)	16/18S Copy Number	Gram Stain
<i>Pseudomonas aeruginosa</i>	B-3509	6.792	1	66.2	4	-
<i>Escherichia coli</i>	B-1109	4.875	1	46.7	7	-
<i>Salmonella enterica</i>	B-4212	4.760	1	52.2	7	-
<i>Lactobacillus fermentum</i>	B-1840	1.905	1	52.4	5	+
<i>Enterococcus faecalis</i>	B-537	2.845	1	37.5	4	+
<i>Staphylococcus aureus</i>	B-41012	2.730	1	32.9	6	+
<i>Listeria monocytogenes</i>	B-33116	2.992	1	38.0	6	+
<i>Bacillus subtilis</i>	B-354	4.045	1	43.9	10	+
<i>Saccharomyces cerevisiae</i>	Y-567	12.1	2	38.3	109 ¹	Yeast
<i>Cryptococcus neoformans</i>	Y-2534	18.9	2	48.3	60 ¹	Yeast

Table 2 continued

Species	NCBI Phylogeny Database
<i>Pseudomonas aeruginosa</i>	Bacteria; Proteobacteria; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas; Pseudomonas aeruginosa group
<i>Escherichia coli</i>	Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacteriales; Enterobacteriaceae; Escherichia
<i>Salmonella enterica</i>	Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacteriales; Enterobacteriaceae; Salmonella
<i>Lactobacillus fermentum</i>	Bacteria; Firmicutes; Bacilli; Lactobacillales; Lactobacillaceae; Lactobacillus
<i>Enterococcus faecalis</i>	Bacteria; Firmicutes; Bacilli; Lactobacillales; Enterococcaceae; Enterococcus
<i>Staphylococcus aureus</i>	Bacteria; Firmicutes; Bacilli; Bacillales; Staphylococcaceae; Staphylococcus
<i>Listeria monocytogenes</i>	Bacteria; Firmicutes; Bacilli; Bacillales; Listeriaceae; Listeria
<i>Bacillus subtilis</i>	Bacteria; Firmicutes; Bacilli; Bacillales; Bacillaceae; Bacillus; Bacillus subtilis group
<i>Saccharomyces cerevisiae</i>	Eukaryota; Opisthokonta; Fungi; Dikarya; Ascomycota; saccharomyceta; Saccharomycotina; Saccharomycetes; Saccharomycetales; Saccharomycetaceae; Saccharomyces
<i>Cryptococcus neoformans</i>	Eukaryota; Opisthokonta; Fungi; Dikarya; Basidiomycota; Agaricomycotina; Tremellomycetes; Tremellales; Tremellaceae; Filobasidiella; Filobasidiella/Cryptococcus neoformans species complex

Protocol

1. Thaw the standard on ice. After thawing, vortex and spin down quickly.
2. The amount of DNA used depends on the library preparation process being evaluated. Example quantities are shown below.

Table 3: Typical DNA input for different library preparation processes¹

Lib. Prep	16S Library	Illumina Nextera® XT	Kapa HyperPlus
DNA input (ng)	10	1	1-2000

Notes:

¹ The table was prepared in April of 2016.

Bioinformatics Analysis Recommendations

1. Assessing accuracy of taxonomy identification

A fundamental goal in microbiome studies is to identify what microbes are present in a sample. After analyzing this microbiome standard using a workflow that includes wet-lab processing and dry-lab interpretation, the taxa identified can be compared with the taxonomy information of the ten strains included in the standard (Table 2, Page 3). This allows a performance assessment of a workflow regarding the limit of the taxonomy resolution, false positives, and false negatives. False positives can be caused by contaminations from wet-lab processes, chimeric sequences during library prep, sequencing errors, demultiplexing errors and defects in bioinformatics analysis. We certify that the impurity level of the standard is <0.01% (by DNA abundance). Therefore, it can be concluded that any alien taxa present at >0.01% (by DNA abundance) in the standard is introduced artificially by the user's workflow. The detection limit of a workflow can be easily determined by checking what strains are detected in the microbiome standard as their abundance follows log distribution.

2. Assessing bias in composition measurement

To assess composition bias, compare the composition profile determined by the user's workflow to the defined composition shown in Table 1. Both wet-lab and dry-lab processes can introduce bias. To determine the quality of a wet-lab process, an accurate/unbiased dry-lab analysis method is needed to interpret the sequencing data from the standard. A straightforward and accurate method to infer the microbial composition from sequencing data of our microbiome standard is through direct read-mapping against reference genomes (or against reference 16S&18S sequences in the case of targeted sequencing). The reference sequences of this microbiome standards can be found in the "Specifications" section of the manual (Page 1, Page 1).

Note: Bacterial strains that are phylogenetically distant can potentially share highly similar sequences in their genomes, e.g. ribosomal RNA sequences and conserved single-copy genes. In the process of direct read mapping, the presence of these highly homologous regions can cause reads that are derived from high-abundance microbes to be assigned to low-abundance microbes, resulting in the overestimation of low-abundance microbes in the standard. One way to overcome this issue is to use a mapping tool that can choose to ignore reads that map to more than one genome. Another way to address this problem is to filter these highly conserved sequences from the reference genomes. Please contact us if you need assistance.

Notes:

Appendix A: Additional Strain Information

Species	NRRL Accession NO.	Strain Name ¹
<i>Bacillus subtilis</i>	B-354	<i>Bacillus subtilis</i> (Ehrenberg 1835) Cohn 1872 ATCC 6633=NRRL B-209=NRS-231=PCI 219
<i>Cryptococcus neoformans</i>	Y-2534	<i>Cryptococcus deneoformans</i> T. Boekout & F. Hagen (2014) 32045=ATCC 32719=CBS 132=CCRC 20528=CCY 17-1-2=DBVPG 6010=IFO 0608=IGC 3957=NRRL Y-8347=PYCC 3957
<i>Enterococcus faecalis</i>	B-537	<i>Enterococcus faecalis</i> (Andrewes and Horder 1906) Schleifer and Kilpper-Bälz 1984 ATCC 7080
<i>Escherichia coli</i>	B-1109	Castellani and Chalmers 1919, 01485cm
<i>Lactobacillus fermentum</i>	B-1840	<i>Lactobacillus fermentum</i> Beijerinck 1901 19lc3=ATCC 14931=BCRC 12190=CCUG 30138=CECT 4007=CIP 102980=DSM 20052=IFO 15885=JCM 1173=KCTC 3112=LMG 6902=NBRC 15885=NCDO 1750=NCIMB 11840=NRIC 1752=NRRL B-4524.
<i>Listeria monocytogenes</i>	B-33116	<i>Listeria monocytogenes</i> (Murray et al. 1926) Pirie 1940 2847=ATCC 19117
<i>Pseudomonas aeruginosa</i>	B-3509	<i>Pseudomonas aeruginosa</i> (Schroeter 1872) Migula 1900 ATCC 15442=NCIB 10421=Pdd-10
<i>Saccharomyces cerevisiae</i>	Y-567	<i>Saccharomyces cerevisiae</i> Meyen ex E. C. Hansen (1883) ATCC 9763=CBS 2978=CBS 5900=CCY 21-4-48=CCY 21-4-54=NCTC 10716=NCTC 7239=NCYC 87=Pattee 6=PCI M-50
<i>Salmonella enterica</i>	B-4212	<i>Salmonella enterica</i> subspecies <i>enterica</i> , Castellani and Chalmers 1919, TA1536
<i>Staphylococcus aureus</i>	B-41012	<i>Staphylococcus aureus</i> Rosenbach 1884

¹ The strain information was extracted from the website of the Agricultural Research Service Culture Collection (NRRL, <https://nrml.ncaur.usda.gov/>).

Ordering Information

Product Description	Size	Catalog No.
ZymoBIOMICS™ Microbial Community DNA Standard II (Log Distribution)	200 ng / 20 µl	D6311

Related Products

Product Description	Size	Catalog No.
Human HCT116 DKO Non-Methylated DNA	5000 ng / 20 µl	D5014-1
ZymoBIOMICS™ Microbial Community Standard II (Log Distribution)	10 preps	D6310
ZymoBIOMICS™ Microbial Community DNA Standard (200ng)	200 ng / 20 µl	D6305
ZymoBIOMICS™ Microbial Community DNA Standard (2000ng)	2000 ng / 20 µl	D6306
ZymoBIOMICS™ Microbial Community Standard	10 preps	D6300
ZymoBIOMICS™ DNA Miniprep	50 preps	D4300

Sample Collection	Size	Catalog No.
DNA/RNA Shield™ – Lysis Tube (Microbe)	50 preps	R1103
DNA/RNA Shield™ – Fecal Collection Tube	10 preps	R1101
DNA/RNA Shield Collection Tube w/ Swab	50 preps	R1107
DNA/RNA Shield™	50 ml	R1100-50
	250 ml	R1100-250
DNA/RNA Shield™ (2X concentrate)	25 ml	R1200-25
	125 ml	R1200-125

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Notes:

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